

Thus the intensity of the resonant microwave will not be constant.

- 7 (a) (i) Minimum energy required for electron to be released from metal surface
(ii) The minimum potential difference such that no electron reaches the collector plate
- (b) (i) Aluminium oxide/printed label blocks the light
(ii) 1. Electrons are released from the surface of can, strips are more positive and repels further
2. Absorbs the beta particles
Strips become less positive and the strips move closer to each other
3. Electrons are not emitted
Strips do not move
- (iii) High intensity means more photons per unit time.
Energy of each photon remains the same
One electron can only absorb energy from one photon at an instance

(c) (i) $\lambda = \frac{hc}{E}$

$$\lambda = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{5000(1.60 \times 10^{-19})}$$

$$= 2.48 \times 10^{-10}$$

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$2.48 \times 10^{-10} = \frac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(1.60 \times 10^{-19})}V}$$

$$V = 24.4 \text{ V}$$

(ii) Conduct experiment in a vacuum

Accelerate the electron through a high potential difference

Pass electrons through a carbon film on to luminous screen

Concentric bright rings

Forms the interference pattern which is a wave property

8 (a) Total initial momentum of a system before collision equals to the total final momentum of the system after collision,

When there is no net external force acting on the system

(b) (i) 1.
$$\text{force of neutron on atom} \left(= \frac{M(\Delta v_{\text{atom}})}{t} \right) = \frac{M(V - 0)}{t}$$